

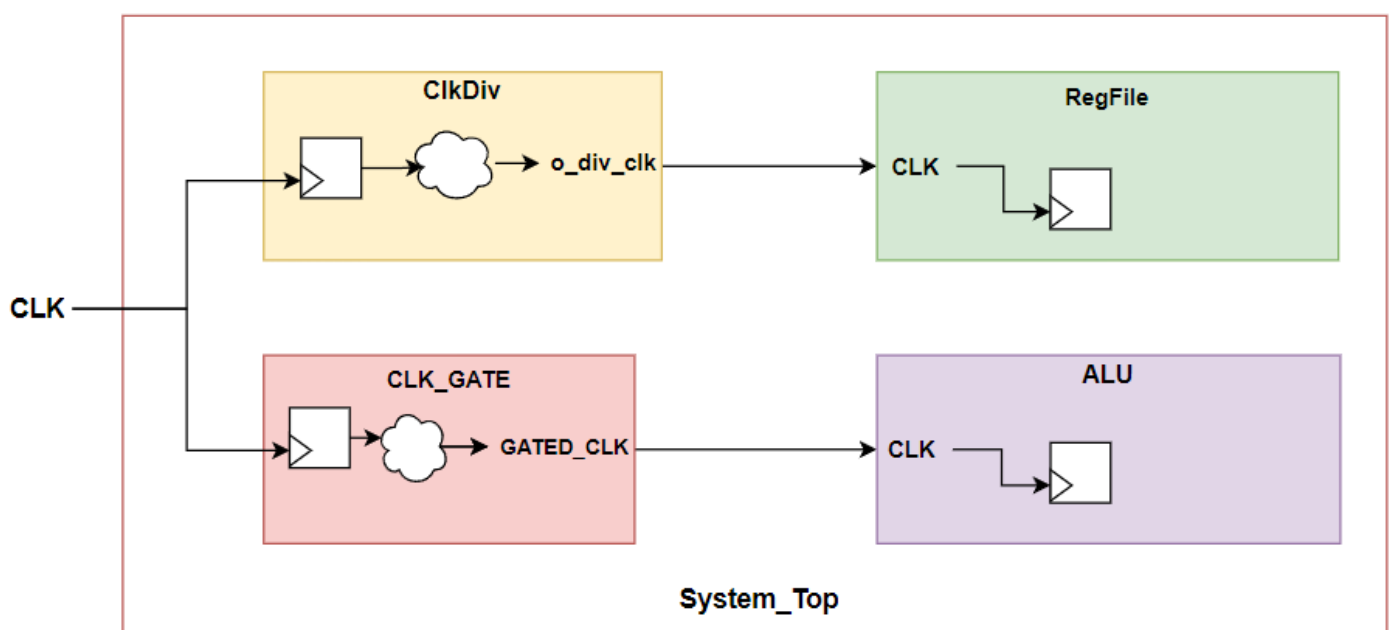
Lab_Syn_3.0

You have to synthesize **System_Top** hierarchy files using **Design Compiler** shell and generate **Technology Dependent** gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside **std_cells** folder:-

- scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db
- scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c
- scmetro_tsmc_cl013g_rvt_ss_1p08v_125c

Steps: -

1. Copy **Lab_Syn_3.0** folder from your computer into the virtual machine inside **Labs** directory
 2. Open a terminal inside path **/home/IC/Labs/Lab_Syn_3.0/syn** and run **syn_script.tcl** script through design compiler shell using the following command in the terminal
 - **dc_shell -f syn_script.tcl | tee syn.log**
 3. Check the synthesis log (**syn.log**) to be sure that: -
 - 1) No Errors
 - 2) No Critical Warnings
 - 3) No Latches
 - 4) No Comb Loops
- Here is a simple block diagram for **System_Top** which represent master clocks as well generated clocks connections to help you to constraint our system



4. Open `syn_script.tcl` file and Add the following constraints in the constraints section before compile command.
 - Create a clock on system clock port with frequency 20 MHz
 - Create generated clock on register file clock with ratio equals 2
 - Create generated clock on ALU clock
 - Create clock uncertainty with 0.2 ns for setup for all master and generated clocks
 - Create clock uncertainty with 0.1 ns for hold for all master and generated clocks
 - Create a clock transition with 0.05 ns for all master and generated clocks
 - Input delays on all input ports except (CLK & RST) with 20% clock period
 - output delays on all output ports with 20% clock period
 - Add Buffer driving cell for all input ports
 - Add load of 75 pf on all output ports
 - Add dont_touch attribute on CLK & RST ports
 - Set operation condition using slow and fast libraries
 - Set wireload model using slow library
 - Add multicycle path constraints on ALU CLK & Register File CLK (Next Session)
5. Report the following reports and redirect each report to a file
 - 1) Power report using (`report_power -hierarchy`) command
 - 2) Area report using (`report_area -hierarchy`) command
 - 3) Setup timing analysis report (`report_timing -max_paths 100 -delay_type max`) command
 - 4) Hold timing analysis report (`report_timing -max_paths 100 -delay_type min`) command
 - 5) Clocks report using (`report_clock -attributes`) command
 - 6) Constraints report using (`report_constraint -all_violators`) command
6. Generate the following files: -
 - 1) Gate level netlist in Verilog format
 - 2) Gate level netlist in ddc format
 - 3) Standard delay format (SDF) file
 - 4) Synopsys design constraints (SDC) file
7. Open Design Compiler GUI, View the Design hierarchy.